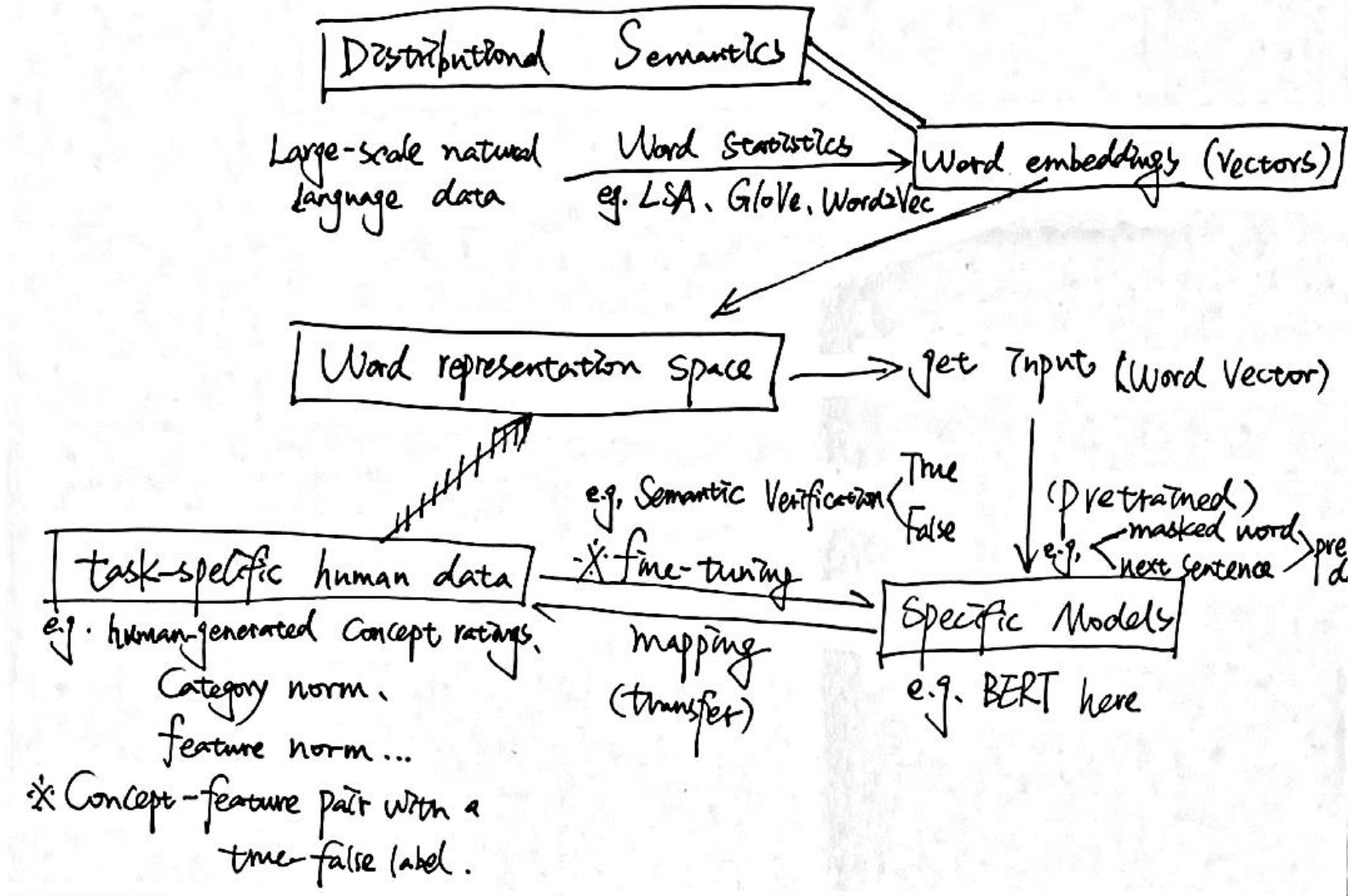
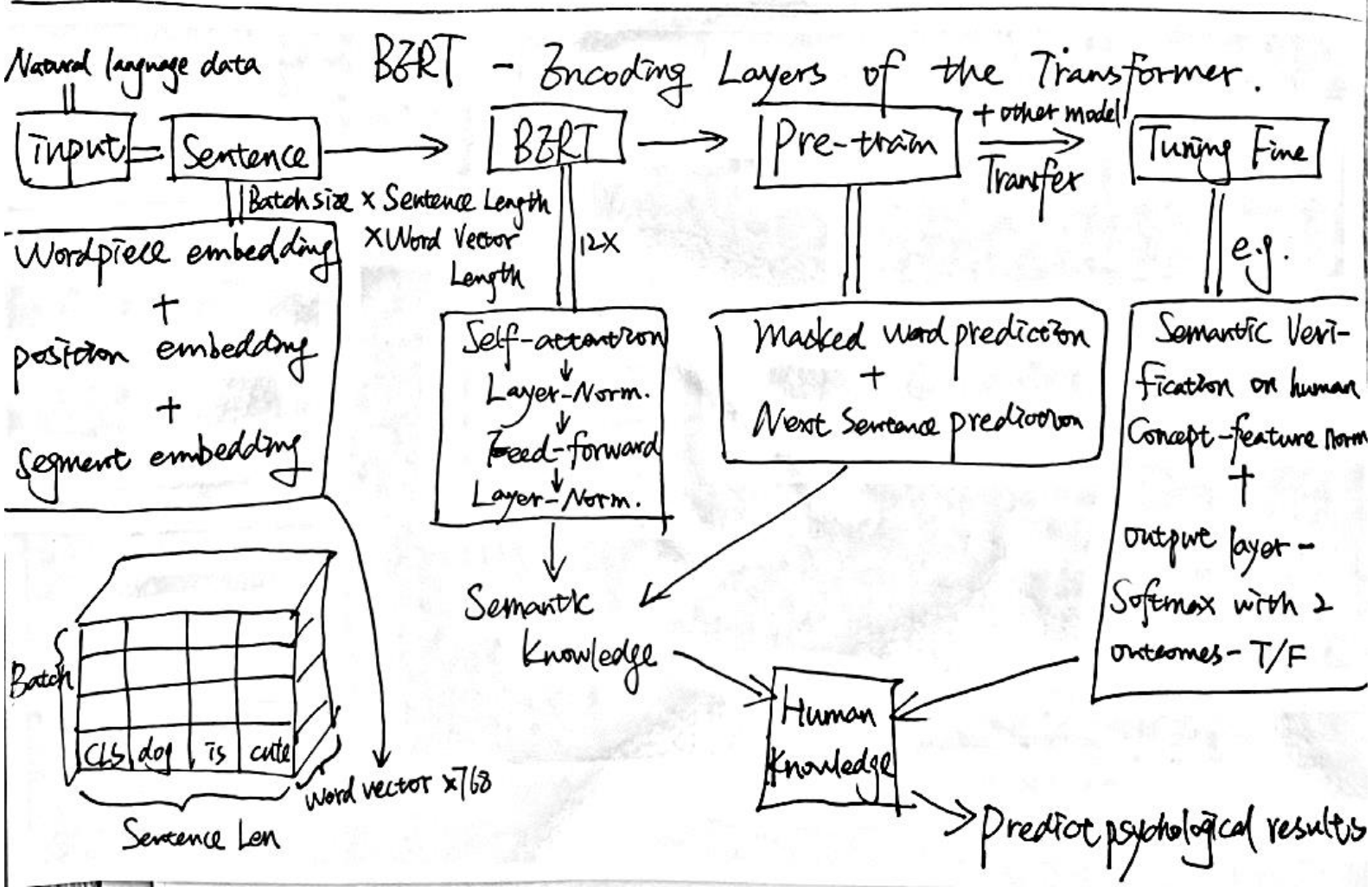
Bhatia, 2022


Distributional Semantics
Large-scale natural language data
Word statistics
eg. LSA, GloVe, Word2Vec
Word embeddings (Vectors)
Word representation space
get Input (Word Vector)
(Pretrained)
e.g. masked word
next sentence
pre
e.g. Semantic Verification
True
False
task-specific human data
fine-tuning
Specific Models
e.g. BERT here
mapping
(transfer)
e.g. human-generated concept ratings.
Category norm.
feature norm...
Concept-feature pair with a true-false label.


BERT - Encoding Layers of the Transformer.
Natural language data
Input
Sentence
BERT
Pre-train
+ other model
Transfer
Tuning Fine
Batch size × Sentence Length × Word Vector Length
12X
e.g.
Wordpiece embedding
+
position embedding
+
segment embedding
Self-attention
Layer-Norm.
Feed-forward
Layer-Norm.
Masked word prediction
+
Next Sentence prediction
Semantic Verification on human Concept-feature norm
+
output layer - Softmax with 2 outcomes - T/F
Semantic Knowledge
Human Knowledge
Predict psychological results
Batch
CLS
dog
is
cute
Word vector ×768
Sentence Len